

Niharika Aggarwal

GenAI / AI Engineer

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📍 Paderborn, Germany (open to relocate)

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🇪🇺 EU work permit; available immediately



WORK EXPERIENCE

Student Assistant: GenAI / Applied AI | *Fraunhofer IEM* | Paderborn, Germany

02/2025 – Present

- Dockerized and deployed multiple Fraunhofer GenAI applications on internal VM infrastructure, enabling reproducible setup, shared access, and production-near evaluation.
- Owned the retrieval pipeline of a RAG-based Product Intelligence demonstrator using Haystack, Python, and MongoDB; reduced information lookup time from approx. 8–10 minutes to under 30 seconds per query.
- Built a full-stack GenAI application with React, Vite, Tailwind, FastAPI, and MongoDB, integrating streaming LLM responses and English/German support; reduced time-to-first-token from approx. 6s to under 1.5s.
- Developed a scalable AI-powered Slide Retrieval system with Qdrant, SQLite, HyDE, vLLM, RabbitMQ, Celery workers, and multilingual OCR; owned retrieval, reranking, and ingestion optimization.
- Improved retrieval quality and ingestion speed through cross-encoder reranking, deduplication, parallelized OCR, and vLLM inference; reduced ingestion time from approx. 45 to 15 minutes per 100-slide deck.
- Developed an enterprise response-generation prototype for Bausch+Ströbel using BAML and Ollama, translating stakeholder requirements into working demos across 2-week sprint cycles.

Working Student: AI / NLP | *safety innovation center gGmbH* | Paderborn, Germany

08/2024 – 01/2025

- Built LLM-based PDF-to-JSON extraction tools to convert unstructured safety and regulatory documents into structured, training-ready datasets, halving manual annotation effort for the research team.
- Fine-tuned open-source LLMs for domain-specific use cases in disaster management, road construction, and traffic regulation, supporting critical infrastructure and emergency-response research projects.
- Designed a LlamaIndex-based RAG pipeline in Node-RED with optimized prompts and curated trusted sources, raising answer relevance scores in internal evaluation from approx. 3.1/5 to 4.3/5.
- Applied NLP techniques (chunking, embedding, cleaning, deduplication) to prepare high-quality training and evaluation datasets across multiple domains.
- Worked cross-functionally with engineers and domain experts to translate real-world safety challenges into deployable AI tooling.

Research Assistant | *Heinz Nixdorf Institute* | Paderborn, Germany

07/2023 – 09/2023

- Contributed to a transfer project building a model-recommendation system (iQBuddy) for Model-Based Systems Engineering (MBSE); led UI/UX design and prototyping.
- Curated a labelled dataset of 200+ entries and applied prompt engineering with H2OGPT and ChatGPT-3 to train and evaluate the recommendation logic, measurably improving output relevance.

Research Intern | *Defence Research and Development Organization (DRDO)* | New Delhi, India

06/2020 – 07/2020

- Conducted research on classical machine learning techniques for predictive analytics, contributing to the Coronavirus Prediction System project under the guidance of a senior scientist.
- Applied cutting-edge ML methodologies (regression, time-series) to health data, building and evaluating predictive models to support pandemic forecasting and decision-making.
- Collaborated in a 2-member intern team, owning end-to-end research tasks from data preprocessing to model evaluation while presenting findings to the supervising scientist.

PROJECTS

Master’s Thesis: GenAI Framework for Analysis of EAM for Prescriptive Analytics 05/2025 – 10/2025
in Manufacturing | *Fraunhofer IEM* | Grade : 1,7

- Designed and implemented Gen-EAMlytics, a GenAI framework for analysing ArchiMate-based EAM models of IoT manufacturing environments and identifying dependencies, synergies, and prescriptive analytics use cases.
- Built a Graph RAG pipeline with Neo4j, FastAPI, React, TypeScript, and OpenAI API, translating natural-language questions into schema-aware Cypher queries grounded on factual knowledge graph data.
- Dockerized and deployed the application on secured VM infrastructure, enabling reproducible setup, remote stakeholder access, and evaluation with six practitioners in a 20-day IoT Factory case study.

Master’s Project: Artificial Intelligence for Systems Engineering (AI4SE) | *Fraunhofer IEM* 10/2023 – 10/2024

- Served as Scrum Master for an 18-person team, leading sprint planning, stakeholder communication, and delivery coordination across a 12-month GenAI project.
- Built a LangChain-based GenAI requirements-generation system using LLMs, RAG, prompt orchestration, and PlantUML generation; reduced token usage by approx. 15% through context optimization.
- Used Paderborn University’s HPC infrastructure for compute-intensive GenAI experimentation, while owning React/Streamlit UI workflows and contributing to the FastAPI backend.

Online Video Streaming Application (Twitch Clone) | *BPIT* 08/2020 - 03/2021

- Developed a streaming web application using HTML, CSS, JavaScript, Node.js, Express, MongoDB with features including live video streaming and chat.
- Awarded 2nd Best University Project.

EDUCATION

Universität Paderborn Paderborn, Germany
M.Sc. Computer Science | Focus: Artificial Intelligence, Software Engineering 10/2022 – 09/2026 expected

- Current grade: 2,0 (*gut / good*)

Guru Gobind Singh Indraprastha University, BPIT New Delhi, India
B.Tech. Information Technology 08/2017 – 07/2021

- Final grade: 1,6 German equivalent (*sehr gut / very good*)

SKILLS

Programming Languages: Python, JavaScript, TypeScript, Java, SQL, HTML, CSS
GenAI & LLM Frameworks: LangChain, LangGraph, LlamaIndex, Haystack, Hugging Face Transformers, BAML, Ollama, vLLM, Azure OpenAI, Google AI Studio
RAG, Agentic AI & LLM Concepts: RAG, Graph RAG, HyDE, Reranking, Tool Calling, Agentic AI, MCP, Prompt Engineering, LLM Fine-tuning (LoRA/QLoRA), RAG Evaluation (RAGAS), NLP, OCR
Backend, APIs & Automation: FastAPI, Node.js, Express.js, REST APIs, API Integration, n8n, Celery, RabbitMQ, Streaming LLM Responses, Workflow Automation
Frontend & AI Prototyping: React, Vite.js, TypeScript, Streamlit, Tailwind CSS, Material UI, Figma, Cursor, GitHub Copilot
Databases & Vector Stores: Qdrant, Chroma, MongoDB, Neo4j, MySQL, SQLite, Firebase
ML, Data Analytics & Visualization: PyTorch, scikit-learn, Pandas, NumPy, Data Cleaning, Data Preparation, Data Visualization, Dashboards, Excel
MLOps, Cloud & Infrastructure: Docker, VM Deployment, Git, GitHub, GitLab, Linux, CI/CD Basics, HPC/Supercomputing Environments
Collaboration, Training & Documentation: Jira, Confluence, Scrum, Kanban, Sprint Planning, AI Training, Workshop Support, MS PowerPoint, L^AT_EX, Markdown

LANGUAGES

English C1: fluent (professional working)
German B1 (Goethe certified) actively progressing toward B2
Hindi Native